$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathcal{S}}(t) = -\mathrm{i}\big[\hat{H}_{\mathcal{S}} + \frac{1}{2}M\widetilde{\Omega}^2(t)\hat{X}^2, \hat{\rho}_{\mathcal{S}}(t)\big] - \mathrm{i}\gamma(t)\big[\hat{X}, \big\{\hat{P}, \hat{\rho}_{\mathcal{S}}(t)\big\}\big] \\ - D(t)\big[\hat{X}, \big[\hat{X}, \hat{\rho}_{\mathcal{S}}(t)\big]\big] - f(t)\big[\hat{X}, \big[\hat{P}, \hat{\rho}_{\mathcal{S}}(t)\big]\big], \quad (5.84)$$